

Q2.

(a).

$$\frac{C(s)}{R(s)} = \frac{K_p}{(s+2)(0.5s+3) + K_p} = \frac{2K_p}{(s+2)(s+6) + 2K_p} \quad \frac{K_p}{6} = 99$$

$$e_{ss} = \frac{K_p}{6+K_p} \quad e_{ss} = \frac{6}{6+K_p} = 1\% \Rightarrow K_p = 6 \times 99 = 594$$